Graph Creation from Relations

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1 Introduction

Though we may not be aware of it, the data we work with for data analytics or machine learning use cases is often relational in nature. This means that the data we perform analysis or learning on is often a set of domaian entitites interacting with each other. Typically, a timestamp for the interaction is typically available.For example, in a retail dataset, we might have customers, products, and transactions, where each transaction links a customer to a product. Constructing a graph from such relational data can help us better understand the relationships and interactions between these entities. Operations Research (Network Models), Machine Learning (Graph Signal Processing, Graphical Models), and Data Science (Graph Analytics) all offer several canonical methods for working with graphs. Creating a graph given a set of relations puts us in a position to explore these methods and apply them to our data. The theory and practice of solving these canonical problems is well established, and if we can construct a graph from our data, we can apply these methods to gain insights, understand the characteristics of the data, and even build predictive models. If you are working with relational data, then there is a good chance that a graph perspective can benefit use case.

There is more to applying the above ideas than simply creating a graph from the data. In the context of machine learning, when working with large datasets, there is typically *heterogeneity* in the relationships between entities. In a supervised learning setting, this may manifest in how predictors are related to each other, and to the target variable. In an unsupervised learning setting, this may manifest in how the data points are related to each other. In either case, the relationships between entities can be complex and multi-faceted.

Typically, this heterogeneity can be characterized by using small number of graphs to represent our data. Each graph has the same predictor response relationship and the same relationship between predictors. In otherwords, rather than using a single graph to represent the entire dataset, we often have to use multiple graphs that are each homogeneous with reference the analysis or learning task.

Heterogeneity has many faces. You can have one analysis task that is heterogeneous in how it manifests in the data or you may want to perform multiple analysis tasks on the same data. For example, you can develop recommendation model from customer transactions, and a clustering model to segment customers. Each of these tasks has a different graph representation of the same data. In summary, heterogeneity can manifest in the need for different models for different parts of the data for the same task, or, for different tasks on the same data. We will see examples of both these cases in the next section.

2 Graph Creation from Relations

I will provide two examples of how to create a graph from relations. In both these examples, I will point out the two points made in the previous section: (1) The data is a set of relations between entities. (2) The data can be viewed as a set of graphs.

The first example is the Olist Case Study. In this example the relations are between customers, products, and purchases(invoices). The description shows two summary perspectives of the data: (1) Temporal perspective, where the data is summarized by week of the year for the year 2017. (2) Spatial perspective, where the data is summarized by state in Brazil. This is an example of heterogeneity arising from different tasks (perspectives) on the same data. The summary shows that the data is relational in nature, and that it can be viewed as a set of graphs. The temporal perspective can be viewed as a graph where each week is a node, and the edges represent the relationships between weeks. The spatial perspective can be viewed as a graph where each state is a node, and the edges represent the relationships between cities in the state. The combination of these two perspectives can be viewed as a graph where a graph is produced for each state and each week, the spatio-temporal graph, can also be produced, but not done in this example. Graph signal processing methods can be applied to analyze, decompose, and summarize a signal such as the revenue on the nodes of the graph. The graph can also be used to build predictive models, such as forecasting revenue for a given week in a given state. The latter (predictive modeling) is not a descriptive task, but it builds on the structure developed in the descriptive task for predictive modeling. Both of these tasks are not done in the example. I will probably do a follow up post on this example to show how to apply graph signal processing methods to the data.

Some discusion of the point about heterogeneity in models for the same task is in order. For the spatial perspective, the data is summarized by city in Sao Paulo state. The cosine similarity in sales is used to create a graph. Cities with similar sales are connected by an edge. The threshold used here is a simple threshold, but a range of techniques can be used to create the graph. Passing the similarity though an affinity kernel is one such technique. Basically, the similarities associated each city are passed through a kernel function to create a graph. Another approach is to use fixed neighborhood size, for example, five nearest neighbors. As a result of appplying a suitable technique to select the edges associated wieh each city, we create a graph that captures the relationships between cities in Sao Paulo state. The connected components of this graph provides the heterogeneity in the relationships between cities. Some cities are not connected to any other city, while others are connected to several other cities. The ones not connected to any city are singleton connected components, while the ones connected to several other cities are larger connected components. The singleton connected components as a group can be analyzed using methods suitable for analyzing Independent and Identically Distributed (IID) data, while the larger connected components can be analyzed using methods suitable for analyzing graphs. *As a consequence, we have two different analysis approaches to understand customer purchasing behavior in Sao Paulo state.* One for the IID data and one for the connected graph data.

The ideas used to identify heterogeneity in the relationships between entities is application and data dependent. There is no one size fits all approach to identifying heterogeneity in the relationships between entities. The key is to use domain knowledge and data analysis techniques to identify a technique for heterogeniety identification. For machine learning problems, there are some common ideas. For unsupervised learning problem, we can leverage covariance or similarity to identify heterogeneity. This is the case for the Olist example. For supervised learning problems, some parts of the data are easier to predict than others. For example, we can have regions of the data that are associated with a single label. These regions have a simple classification rule. Other regions may require more sophisticated classifiers. These other regions can be profiled by a By adjusting the height of the tree and checking that results generalize, the dataset can be partitioned into regions (the leaves of the tree) that are each homogeneous from the standpoint of the learning task. For more details, please refer to [1].

The second example is the

References

[1] Rajiv Sambasivan and Sourish Das. Classification and regression using augmented trees. *Int. J. Data Sci. Anal.*, 7(4):259–276, 2019.